



Pharm*Achieve*

PHARMACY MANAGEMENT MODULE 2: FORMULARY, BILLING, AND SERVICES

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

CDA-AMC Canada's Drug Agency- L'Agence des Médicaments du Canada

BCGP Board Certified Geriatric Pharmacist

CDE Certified Diabetes Educator

DUE Drug Use Evaluation

EAP Exceptional Access Program

HTA Health Technology Assessment

LU Limited Use Code

MAC Medical Advisory Committee

MD Medical Doctor

NSAID Nonsteroidal Anti-inflammatory Drug

ODB Ontario Drug Benefit

OFI Off Formulary Interchangeability

P/T Provinces (or Provincial) and Territories (or Territorial)

P & T Pharmacy and Therapeutics

PBM Pharmacy Benefit Manager

RCMP Royal Canadian Mounted Police

RN Registered Nurse

Rx Prescription

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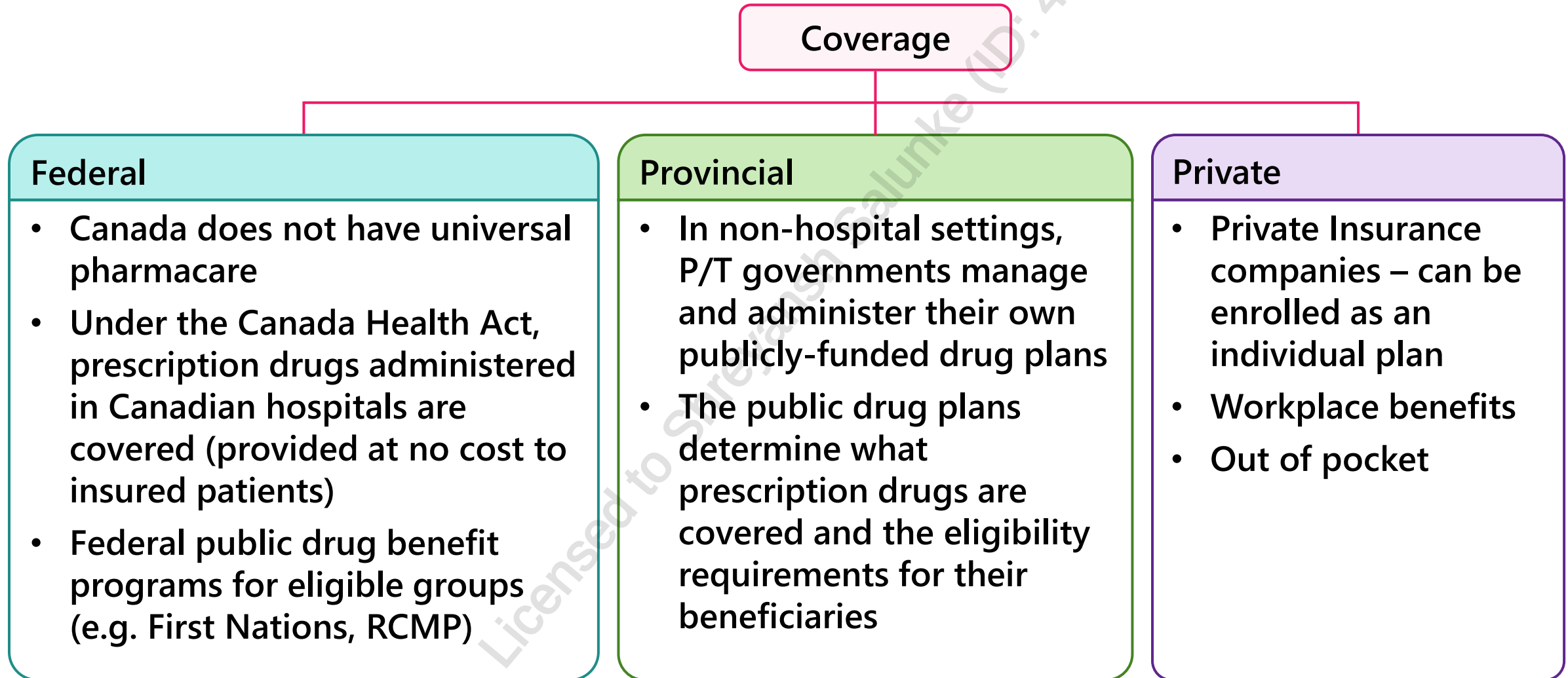
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PRESCRIPTION DRUG COVERAGE IN CANADA



PROVINCIAL DRUG PLANS

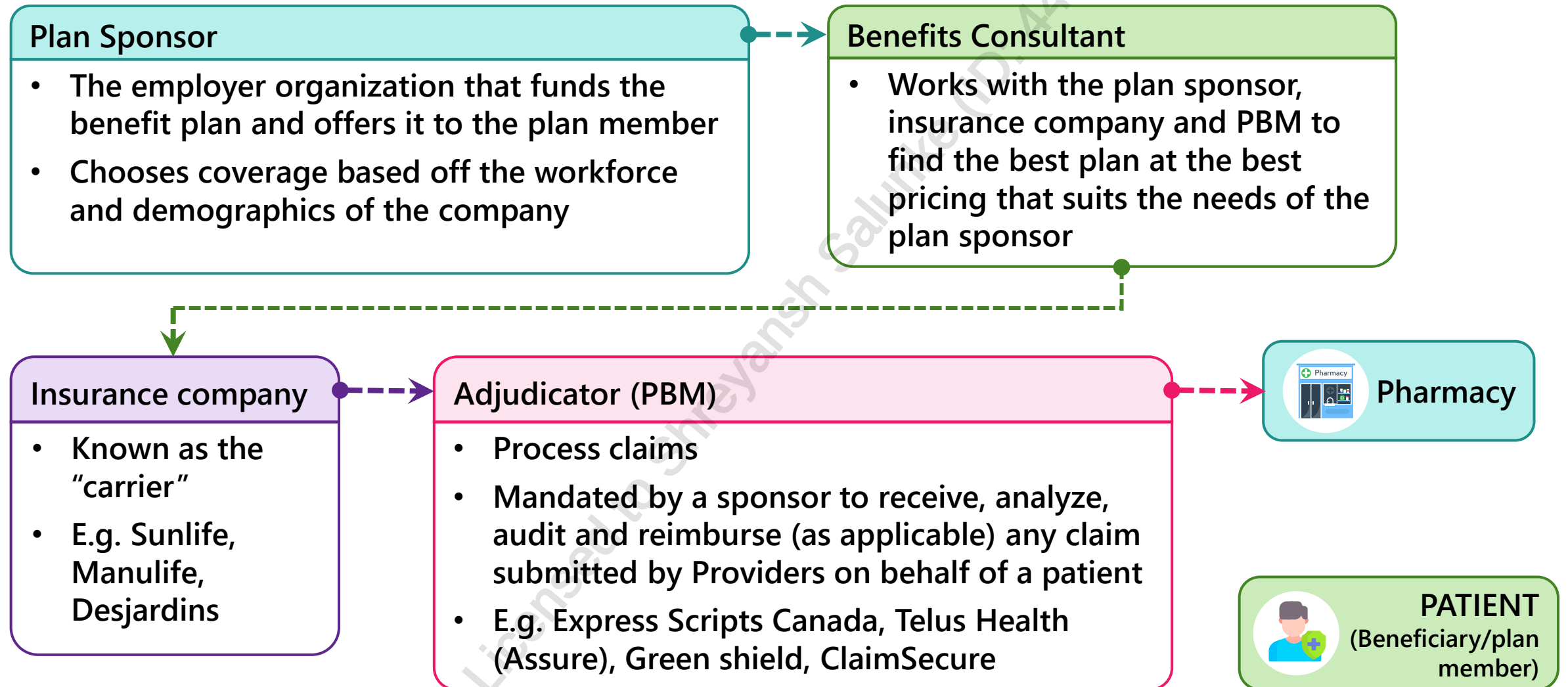
- Each provincial and territorial government offers a drug benefit plan for eligible groups
 - Income-based programs
 - Specialized coverage for specific patient groups (e.g. seniors, recipients of social assistance, high drug costs)

FYI

- Alberta: Prescription Drug Programs
- British Columbia: Pharmacare
- Manitoba: Pharmacare Program
- New Brunswick: Prescription Drug Program
- Newfoundland: Pharmaceutical Services
- Northwest Territories: Extended Health Benefits Program
- Nova Scotia: Pharmacare
- Nunavut: Nunavut Health Care Plan
- Ontario: Drug Benefit Program
- Prince Edward Island: Drug Cost Assistance Programs
- Quebec: Prescription Drug Insurance
- Saskatchewan: Drug Plan
- Yukon: Pharmacare Program



PRIVATE DRUG PLANS



DRUG FORMULARY

WHAT IS A DRUG FORMULARY?



A list of therapeutically effective drugs of proven high quality that have been approved for coverage under the Drug Plan.

- E-health Saskatchewan

- Federal, provincial and/or territorial agencies managing drug prices will conduct a review to decide if the new medication will be included into the public drug plans and under what terms
- When conducting a Health Technology Assessment (HTA), *the following must be addressed*:
 1. Does it work? How does it compare to available products on the market? *Efficacy*
 2. At what cost? *Financial, safety*
 3. Is it worth it?
- Example of Health Technology Assessment Agency in Canada is CDA (Canada's Drug Agency, formerly known as CADTH - Canadian Agency for Drugs and Technologies in Health)
- Private plans conduct their own review and HTA

DRUG FORMULARY

ONTARIO DRUG BENEFIT FORMULARY EXAMPLE

Formulary Search

Ontario

Search the **Ontario Drug Benefit Formulary/Comparative Drug Index**, effective from **July 31, 2024** using any or all of the criteria below.

Coverage Status

All Benefits

Therapeutic Classification

All Therapeutic Classifications

Manufacturer

All Manufacturers

Keyword

Keyword Type

☒ Generic Name

☐ Brand Name

☐ DIN/PIN/NPN

☐ Search for Products that begin with Keyword entered

☐ Generic/Brand name Summary List

Search

Reset

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DRUG FORMULARY

ONTARIO DRUG BENEFIT FORMULARY RIVAROXABAN EXAMPLE

DIN/ PIN/ NPN ◇	Generic Name ◇	Brand Name, Strength & Dosage Form ◇	MFR ◇	Drug Benefit Price or Unit Price ◇	Amount MOH Pays ◇	Inter- change- able ◇	Limited Use ◇	Thera- peutic Notes
02470519	RIVAROXABAN	Apo-Rivaroxaban 20mg Tab	APX	0.7175	0.7175	YES	YES	NO
02470500	RIVAROXABAN	Apo-Rivaroxaban 15mg Tab	APX	0.7175	0.7175	YES	YES	NO
02541734	RIVAROXABAN	Apo-Rivaroxaban 2.5mg Tab	APX	0.3550	0.3550	YES	YES	NO
02470497	RIVAROXABAN	Apo-Rivaroxaban 10mg Tab	APX	0.7175	0.7175	YES	YES	NO
02512076	RIVAROXABAN	PMS-Rivaroxaban 20mg Tab	PMS	0.7175	0.7175	YES	YES	NO
02512068	RIVAROXABAN	PMS-Rivaroxaban 15mg Tab	PMS	0.7175	0.7175	YES	YES	NO
02527537	RIVAROXABAN	PMS-Rivaroxaban 2.5mg Tab	PMS	0.3550	0.3550	YES	YES	NO

INTERCHANGEABILITY

- The formulary contains both brand name and generic drugs
- If there is a generic interchangeable product available under the benefit program, the Ministry of Health and Long-Term Care will reimburse a pharmacist for the **lowest cost interchangeable product** (e.g. a generic)
- Higher-cost interchangeable drugs (e.g. brand name medications) are only covered by the Ministry when deemed medically necessary

DRUG FORMULARY

ONTARIO DRUG BENEFIT FORMULARY RIVAROXABAN EXAMPLE

444	For the treatment of deep vein thrombosis (DVT) or pulmonary embolism (PE) for up to six (6) months.
	LU Authorization Period: 6 Months
	NOTE: -The recommended dose of rivaroxaban for patients initiating DVT or PE treatment is 15 mg twice daily for 3 weeks, followed by 20 mg once daily. -ODB Program coverage for rivaroxaban is an alternative to heparin/warfarin for up to 6 months. When used for greater than 6 months, rivaroxaban is more costly than heparin/warfarin. As such, patients with an intended duration of therapy greater than 6 months should be considered for initiation on heparin/warfarin. -For clarity, coverage will not be provided for patients who have already received 6 months of treatment with apixaban for the same DVT or PE. -Since renal impairment can increase bleeding risk, it is important to monitor renal function regularly. Other factors that increase bleeding risks should also be assessed and monitored (see product monograph).

LIMITED USE (LU) CODE

- LU products are only reimbursed when used for **eligible recipients** and meet **specific clinical criteria**

DRUG FORMULARY

ONTARIO DRUG BENEFIT FORMULARY XARELTO® EXAMPLE

DIN/ PIN/ NPN ◇	Generic Name ◇	Brand Name, Strength & Dosage Form ◇	MFR ◇	Drug Benefit Price or Unit Price ◇	Amount MOH Pays ◇	Inter- change- able ◇	Limited Use ◇	Thera- peutic Notes
02480808	RIVAROXABAN	Xarelto 2.5mg Tab	BAH	1.4200	0.3550	YES	YES	NO
02378612	RIVAROXABAN	Xarelto 20mg Tab	BAH	2.8700	0.7175	YES	YES	NO
02378604	RIVAROXABAN	Xarelto 15mg Tab	BAH	2.8700	0.7175	YES	YES	NO
02316986	RIVAROXABAN	Xarelto 10mg Tab	BAH	2.8700	0.7175	YES	YES	NO

BRAND CARD

- Allows patients to **stay on their brand name drugs at the generic price**
- Provides additional drug coverage on many brand-name medications, and other medical products, including blood glucose meters and test strips
- It has no monthly charges and is valid at most Canadian pharmacies
- Examples: InnoviCares™ and RxHelpONE™

DRUG FORMULARY

ONTARIO DRUG BENEFIT FORMULARY CETIRIZINE EXAMPLE

DIN/ PIN/ NPN ◇	Generic Name ◇	Brand Name, Strength & Dosage Form ◇	MFR ▼	Benefit Price or Unit Price ◇	Amount MOH Pays ◇	Inter- change- able ◇	Limited Use ◇	Thera- peutic Notes
02528681	CETIRIZINE HYDROCHLORIDE	Teva-Cetirizine (Off-Formulary Interchangeable) 20mg Tab	TEV	0.7535	N/A	YES	NO	NO
02534126	CETIRIZINE HYDROCHLORIDE	Cetirizine (Off-Formulary Interchangeable) 20mg Tab	SIV	0.7535	N/A	YES	NO	NO
02515695	CETIRIZINE HYDROCHLORIDE	Cetirizine (Off-Formulary Interchangeable) 20mg Tab	SAI	0.7535	N/A	YES	NO	NO

Off-Formulary Interchangeability (OFI)

- The ministry has a process in place to determine interchangeability for both formulary (products considered as benefits or eligible for reimbursement) and off formulary products
- OFI is the application of interchangeable designations to drug products where the **original (e.g. brand name) products are not listed as benefits in the formulary**
- Some OFI products may be eligible for coverage under ODB through the EAP

DRUG FORMULARY

ONTARIO DRUG BENEFIT FORMULARY OTHER EXAMPLE

Example: Maxalt® (Rizatriptan) - *For the treatment of migraines with or without aura in patients who failed adequate trials of other medications for migraines (e.g. acetaminophen, NSAIDs) and where the following information is provided:*

- Details on prior trial of migraine prophylactic medications (e.g., amitriptyline, beta-blockers), or rationale why they were deemed inappropriate
- Details on the nature of the migraine attacks (i.e. the number, duration, and severity)
- Duration of Approval: 5 years

Exceptional Access Program

- Facilitates patient access to drugs and products **not listed on the Formulary** or where no listed alternative is available
- To apply through the EAP, the patient's prescriber (i.e. physician or nurse practitioner) must submit a request documenting complete and relevant medical information to the ministry, providing the clinical rationale for requesting the unlisted drug and reasons why covered benefits are not suitable
- Coverage durations will vary depending on the drug and the condition and are determined by the funding criteria that have been approved for the drug

DRUG FORMULARY

ONTARIO DRUG BENEFIT FORMULARY OTHER EXAMPLE

Nutrition Products

- Enteral nutrition products are eligible for coverage under the provincial program only when prescribed by a physician or nurse practitioner as the **patient's sole source of nutrition**
- *Patients must meet one of the following criteria:*
 - Oropharyngeal or gastrointestinal disorders resulting in esophageal dysfunction or dysphagia (e.g., head and neck surgery, neuromuscular disorder, or cerebral vascular disease where dysphagia prevents eating)
 - Maldigestion or malabsorption disorder and/or significant gut failure where food is not tolerated; (e.g., pancreatic insufficiency, biliary obstruction, short bowel syndrome)
 - For patients requiring the use of a chemically defined diet as a primary treatment of a disease where the therapeutic benefit has been demonstrated (i.e., Crohn's disease)
- **Example:** PurAmino A+ pediatric formula- hypoallergenic amino acid-based for infants with severe cow's milk allergy

DRUG FORMULARY PRACTICE QUESTIONS

1. A pharmacy professional is looking to add drug Y on the pharmacy formulary. What is the MOST important question to consider?
 - a) Is the drug available in oral and liquid forms?
 - b) Is there a Health Canada warning on a possibly harmful adverse effects?
 - c) Can the manufacturing company provide samples for drug Y and incentives?
 - d) Is the drug interchangeable?

2. You are the pharmacist and receive a prescription for Sandoz-Pantoprazole sodium 40mg, however you look at the patient profile and there is a note to dispense only Mylan-Pantoprazole sodium. How do you proceed?
 - a) Dispense Sandoz-Pantoprazole as per order from the doctor
 - b) Hold the prescription and fax the doctor to clarify
 - c) Ask the patient why they prefer Mylan and dispense Mylan brand
 - d) Ask the patient to fill the prescription at another pharmacy

PROVINCIAL HEALTH NETWORK SYSTEM

The Health Network System is a province-wide computer system that links pharmacies directly to the ministry. In Ontario, it is used to process claims under the **Ontario Drug Benefit (ODB) Program**



Ontario Drug Benefit (ODB) Program

If you qualify for the program, your prescription information and health card number are entered into the Pharmacy computer system (Ex: Kroll, HealthWatch, Nexxsys)

The information is communicated online to the ministry's computer system. This can identify drug interactions, duplicate prescriptions, double doctoring, frauds

Ministry checks your eligibility, identifies if the drug is on the provincial formulary and calculates how much it will pay for the prescription

Pharmacy charges you for your portion of the prescription costs and dispensing fee

KEY TERMINOLOGY

Dispensing Fee

- **Flat fee set by the pharmacy for each drug dispensed** that covers the cost of preparing and dispensing the drug (e.g. counting tablets, labelling, counselling, checking drug interactions)
- Dispensing Fee Caps: drug plans may limit or cap the amount that will be reimbursed for a dispensing fee
 - E.g. The dispensing fee is set to \$11.99, the patient's plan has a dispensing fee cap of \$9.99, thus the patient must pay \$2 out-of-pocket



There are different types of insurance plans with many exceptions. But follow the rules outlined below unless the question instructs you to do otherwise

Copayment ("copay")

- An **upfront fixed fee** that the plan member must pay out-of-pocket for each prescription filled
- E.g. A \$5 copay means the plan member must pay a \$5 copayment for each prescription dispensed

Deductible

- A **fixed amount** to be paid by the plan member out-of-pocket **before the plan will begin to cover** their prescription drug expenses (usually charged annually)
- E.g. If a plan has a deductible of \$500, the plan member must pay the full cost of each prescription until they contribute \$500 towards prescription costs before the plan begins covering their drug expenses

KEY TERMINOLOGY

Coinsurance

- The **percentage of a covered healthcare service (e.g. drug cost)** that the employer or the patient pays **after the deductible is fulfilled**
- May be expressed from the perspective of the insurance company or the patient e.g. if a patient has “10% coinsurance”, this means the plan offers 90% coverage on cost of the drug, and the plan member pays the remaining 10% out of pocket



There are different types of insurance plans with many exceptions. But follow the rules outlined below unless the question instructs you to do otherwise

Co-ordination of Benefits

- A situation in which a **plan member is covered under more than one public and/or private insurance plan** and payment of the member's covered medications must be **coordinated between the plans and their respective payers** to determine the amount payable under each plan
- E.g. a patient is covered by their own private insurance and their spouse's insurance, the prescription is processed through the primary plan first and their spouse's plan will cover the remaining amount

Max Cost Exceeded

- A situation in which a plan member **reaches the maximum amount that their plan will cover** and must **pay for all future expenses out-of-pocket**
- E.g. a plan covers \$1000 per year in prescription cost, once the member spends \$1000 on prescriptions, they start to pay for their own prescription costs

PRESCRIPTION COST

Three factors determine the cost of prescription drugs:

Drug Ingredient Cost

The amount pharmacy pays to buy the drug



Mark-Up

Any additional amount a pharmacy may charge for a drug, above the original drug cost

- The mark-up is applied to help pay for the costs of running the pharmacy



Dispensing Fee

The professional fee a pharmacist charges to fill a prescription

Total Prescription Cost

INSURANCE CALCULATIONS PRACTICE QUESTION

Question 1: MK is a 25-year-old female who is a new patient at your pharmacy. She is looking to fill a prescription for escitalopram 10 mg and is requesting a 30-day supply. She has a private insurance plan that has no deductible, covers prescriptions fully with no dispensing fee cap, but must pay a \$5 co-pay per medication filled.

- The cost of escitalopram is \$9 (for a 30-day supply)
- The pharmacy dispensing fee is \$12 per prescription

a) What is the total cost of MK's prescription if she has no private plan?

b) How much will MK need to pay the pharmacy with her current coverage?

INSURANCE CALCULATIONS PRACTICE QUESTION

Question 1: MK comes back 1 month later to update her new insurance information. She wants to pick up a 60-day supply of escitalopram 10 mg. She tells the pharmacist that she now has a coinsurance of 5% and there is now a dispensing fee cap of \$10. The plan still has no deductible.

- The cost of escitalopram is \$9 (for a 30-day supply)
- The pharmacy dispensing fee is \$12 per prescription

c) How much will MK pay for her prescription?

INSURANCE CALCULATIONS PRACTICE QUESTION

Question 2: DP is filling a new prescription for 30 days and has drug plan A. He currently has \$18 left on his deductible and the new prescription has a drug cost of \$150 for a 30-day supply. This is DP's first fill for this medication.

- The pharmacy dispensing fee is \$12 per prescription

How much will DP need to pay the pharmacy?

DRUG PLAN A:

- \$100 annual deductible
- Covers 80% of drug cost (coinsurance)
- Covers up to \$8 of dispensing fee
- Has maximum of 100 day-supply

INSURANCE CALCULATIONS PRACTICE QUESTION

Question 2: DP is filling a new prescription for 30 days and has drug plan A. He currently has \$18 left on his deductible and the new prescription has a drug cost of \$150 for a 30-day supply. This is DP's first fill for this medication.

- The pharmacy dispensing fee is \$12 per prescription

How much will DP need to pay the pharmacy?

DP first needs to pay the deductible of his drug costs before his plan starts to pay: $\$150 - \$18 = \$132$

DRUG PLAN A:

- \$100 annual deductible
- Covers 80% of drug cost (coinsurance)
- Covers up to \$8 of dispensing fee
- Has maximum of 100 day-supply

With the deductible now having been paid, his plan is activated. His plan pays 80% so DP must pay co-insurance of 20% of the remaining amount; i.e. $20\% \text{ of } \$132 = \26.40

In addition, DP must pay the portion of the dispensing fee not covered by insurance: $\$12 - \$8 = \$4$.

In total DP pays:

$\$48.40 = \$18 \text{ deductible} + \$26.40 \text{ (20\% co-insurance)} + \$4 \text{ (dispensing fee not covered by his plan)}$

INSURANCE CALCULATIONS PRACTICE QUESTION

Question 2- Part 2: DP is filling a new prescription for 30 days and has drug plan B. He has paid off his deductible, and the new prescription has a drug cost of \$250 for a 30-day supply. This is his first fill for this prescription

- The pharmacy dispensing fee is \$12 per prescription

How much will DP need to pay the pharmacy?

DRUG PLAN B:

- \$200 annual deductible
- Has a copay of \$5
- Covers up to \$10 of dispensing fee
- Has maximum of 100 day-supply

INSURANCE CALCULATIONS PRACTICE QUESTION

Question 3 - YS is a 67-year-old male who is a regular patient at your pharmacy. He is requesting you to fill four of his prescriptions for a 30-day supply - rosuvastatin 5 mg for dyslipidemia, telmisartan 80 mg for hypertension, tamsulosin 0.4 mg for benign prostatic hyperplasia and Viagra® 100mg for erectile dysfunction. He has met his deductible for the year and has a copay of \$5. His plan will pay up to a maximum of \$8 of your pharmacy's dispensing fee. Note his drug plan does not cover Viagra®.

- The pharmacy dispensing fee is \$13

What is the total cost of filling all four medications with his drug plan?

The cost of 30-day supply of his medications is as follows:

- Rosuvastatin 5 mg = \$15
- Telmisartan 80 mg: \$10
- Tamsulosin 0.4 mg: \$21
- Viagra® 100mg: \$68



What if he had no drug plan?

INSURANCE CALCULATIONS PRACTICE QUESTION

Question 3 - YS is a 67-year-old male who is a regular patient at your pharmacy. He is requesting you to fill four of his prescriptions for a 30-day supply – rosuvastatin 5 mg for dyslipidemia, telmisartan 80 mg for hypertension, tamsulosin 0.4 mg for benign prostatic hyperplasia and Viagra® 100mg for erectile dysfunction. He has met his deductible for the year and has a copay of \$5. His plan will pay up to a maximum of \$8 of your pharmacy's dispensing fee. Note his drug plan does not cover Viagra®.

- The pharmacy dispensing fee is \$13

What is the total cost of filling all four medications with his drug plan?

- *\$5 copay and \$5 dispensing fee portion will be applied to each of the 3 drugs that are covered by insurance (rosuvastatin, telmisartan and tamsulosin) → so $3 \times (5+5) = \$30$*
- *Viagra® is not covered at all by insurance, so the patient pays drug cost (\$68) + dispensing fee (\$13) → **\$81***
- *In total, patient pays $\$30 + \$81 = \underline{\$111}$*

The cost of 30-day supply of his medications is as follows:

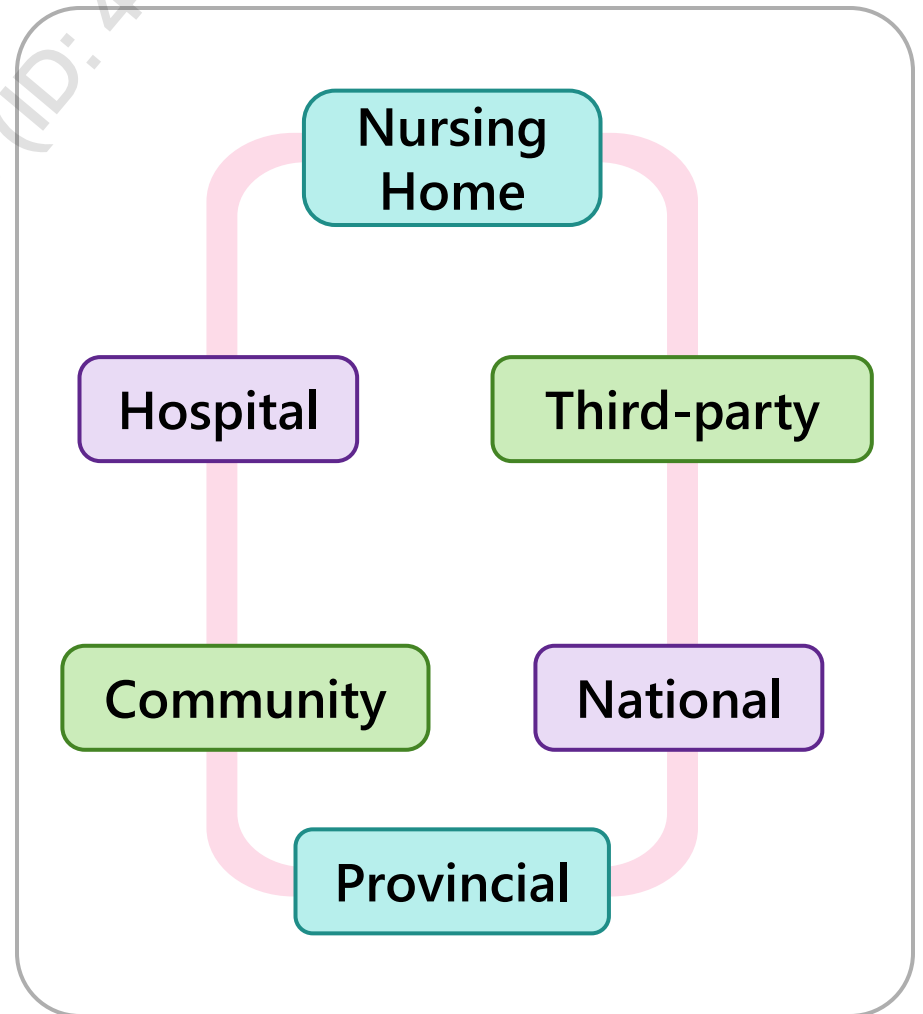
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What if he had no drug plan?

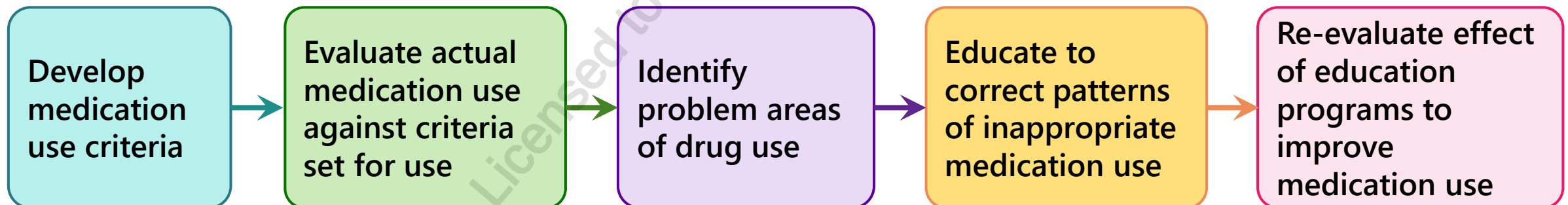
DRUG USE EVALUATION (DUE)

- Performance improvement method that focuses on **evaluating and improving medication-use processes** with the goal of **optimizing patient treatments and enhancing patient safety**
- Used to **promote medication safety**, evaluate if medication is meeting **quality standards** and to determine **the cost-effectiveness** of therapy
- May be applied to a medication, therapeutic class, disease state, medication-use process or specific clinical outcome
- The organizational body (e.g., quality management committee, P & T committee) responsible for the DUE process should include **prescriber, pharmacist, nurse, and administrator**



DRUG USE EVALUATION (DUE)

What is it?	Why is it done?	How is it done?
Criteria based, ongoing, systematic quality assurance process designed to ensure that drugs are used appropriately, safely and effectively	- To measure cost-effectiveness - To capture inappropriate prescribing patterns	- Either retrospectively or prospectively - Actual drug use is compared to standards (how the drug should be used) - Based on findings → can provide prescribing education - PDSA cycle (Plan-Do-Study-Act) is done to determine compliance with standard - PDSA cycle is a quality improvement process to be taken for all initiatives



DRUG USE EVALUATION

NATIONAL AND PROVINCIAL LEVEL

National

- Health Canada approves the **drug for marketing and sale**
- Expert panel, **Canada's Drug Agency (CDA)**, reviews drugs and provides recommendations on whether the drug should be added to the formulary
- Every new marketed novel drug therapy (exception: new generic drugs) needs to be reviewed by CDA prior to submission for provincial formulary

Provincial

- Example: Ontario Drug Benefit (ODB) formulary and the ministry's drug advisory committee, Committee to Evaluate Drugs (CED)
 - ODB reviews CDA recommendations & **makes decisions based on resources available**, need for the drug based on population etc.
 - The provincial body does not have to follow the CDA recommendations as **they make their own decisions** about adding drugs to the provincial formulary

DRUG USE EVALUATION

NURSING HOME AND THIRD PARTY

Nursing home

- Pharmacist consultant working with MD, RNs to review medications & offer suggestions as medication experts
- Academic detailing
 - E.g. pharmacists providing one-on-one education to other HCPs regarding drug use in the elderly

Third party

- Limited list of approved drugs & OTCs
- Coverage driven by employer's benefits (varies)
- Focus is mainly on cost as opposed to clinical evidence

DRUG USE EVALUATION

HOSPITAL

- Hospital formulary can be **restrictive** (provide selected products and substitute drug with a recommended alternative that is available) or **unrestrictive** (provide all medications)
- Formulary decision is based on drug-use evaluation which is reviewed by **Pharmacy and Therapeutics (P&T)** and **approved by Medical Advisory Committee (MAC)**
 - Pharmacy & Therapeutics Committee is a sub-committee of Medical Advisory Committee
 - Acts as link between pharmacy department and other disciplines and health care providers
 - **Mission is to promote safe, rational and cost-effective distribution, utilization and administration of drugs**
 - Maintain a formulary with optimal & cost-effective therapy
 - Develop, review & revise policies and procedures about safe, cost-effective drug use & administration of medications
 - Review use of approved medication for unapproved indications and make recommendations
 - Complete drug utilization review of expensive drugs for **cost-effectiveness**

DRUG USE MANAGEMENT

COMMUNITY

- Primary source of drug use management is through patient monitoring for **efficacy and side effects**
- Ensure that provincial formulary restrictions are followed
- Provinces **compensate pharmacists for collaboration** on drug use management strategies and cognitive services
 - Trial prescriptions, change in prescription dose, change in drug therapy due to intolerance, change in drug therapy due to inappropriate drug, refuse to fill prescription, smoking cessation, MedsChecks, etc.
- Programs to enhance appropriate prescribing
 - E.g. academic detailing to physicians & pharmacists

PROFESSIONAL SERVICES

A *professional pharmacy service* is an action or set of actions undertaken in or organized by a pharmacy, delivered by a pharmacist or other health practitioner, who **applies their specialized health knowledge** personally or via an intermediary, with a patient/client, population or other health professional, to **optimize patient outcomes and process of care**

Common Professional Services

- Medication reviews (Basic and Diabetes-related)
- Diabetes care (CDE)
- Geriatric care (BCGP)
- Pharmaceutical opinion
- Minor Ailment prescribing
- Immunization
- Naloxone kit administration
- Paxlovid® prescribing
- Compliance pack
- Smoking cessation (refer to individual lecture)
- Travel consultations
- Point-of-care testing
- Pharmacist Rx renewals and adaptations



PROFESSIONAL SERVICES

MEDICATION REVIEW



An expert assessment to verify medications, dosage, side effects and potential interactions. It also provides the opportunity to ask questions to better understand medications and the proper way they should be taken, reducing the risk for medication-related problems.

- CPhA

- A medication therapy review is a **one-on-one consultation between a pharmacist and a patient** during which the pharmacist collects information about the patient's health conditions and all the medications he/she takes – whether prescribed, over-the-counter, or a natural health product
 - The pharmacist will check to ensure that each medication is **appropriate, effective and safe and there are no adverse effects or interactions**
 - Patient education is also an important part of the review
 - Pharmacists can identify the most **cost-effective choice** for medications prescribed
 - **Reduces costs to the health plan** because of reduced risk of complications from drug interactions and adherence issues, reduces risk of hospitalization
- Can also be a specialized review such as a **diabetes-oriented medication review**

PROFESSIONAL SERVICES

MEDICATION REVIEW

CASE STUDY

A patient comes to your pharmacy for a medication review. During the interview, he takes out a medication list and shows you (the pharmacist) all the medications he is currently taking:

- Tamsulosin - 0.4mg daily
- Zopiclone - 7.5mg QHS
- Amlodipine - 5mg QAM
- Metformin – 500mg BID
- Quetiapine - 25mg QHS
- Trazodone – 50mg QHS
- Benadryl® over the counter for seasonal allergies

The patient also reveals that he is always feeling “tired and groggy” throughout the day, so he forgets to take the morning medications.

How do you approach this situation?

PROFESSIONAL SERVICES

PHARMACEUTICAL OPINION

"A clinical intervention that occurs at the time of dispensing a prescription or when conducting a MedsCheck review, in which a pharmacist identifies a potential concern that requires consultation with the prescriber. Based on this, the medication may be dispensed as prescribed, it may not be dispensed, or the therapy may be adjusted or changed". - *OPA*



Ontario Pharmacists Association | Pharmaceutical Opinion Program (POP)

PROFESSIONAL SERVICES

MINOR AILMENT PRESCRIBING

Health conditions that can be managed with **minimal treatment and/or self-care strategies**

- Usually a short-term condition
- Lab results aren't usually required
- Low risk of treatment masking underlying conditions
- No red flags that could suggest a more serious condition – if present, a referral is made
- Only minimal or short-term follow-up is required

WHAT ARE MINOR AILMENT EXAMPLES?

- Dysmenorrhea
- Gastroesophageal reflux disease (GERD)
- Hemorrhoids
- Herpes Labialis (cold sores)
- Impetigo
- Insect bites
- Tick bites (post exposure prophylaxis)
- Urinary Tract infections
- Acne
- Allergic rhinitis
- Aphthous ulcers (canker sores)
- Candidal Stomatitis (Oral Thrush)
- Conjunctivitis (Bacterial, viral, allergic)
- Dermatitis (atopic, eczema, allergic, contact)
- Diaper dermatitis
- Nausea and vomiting of pregnancy

PROFESSIONAL SERVICES

IMMUNIZATION

CASE STUDY

You are busy administering influenza vaccines in your clinic and are alerted to the fact that a patient of yours appears to be in distress following vaccination.

- What are the signs and symptoms of anaphylaxis?
- How does this differ from fainting?
- How do you manage it?

SIGNS OF ANAPHYLAXIS

- Urticaria, itchiness, angioedema (e.g. swollen lips, tongue, or throat)
- Respiratory distress (e.g. wheezing, shortness of breath, stridor)
- This usually occurs minutes or up to 4 hours after vaccination



PROFESSIONAL SERVICES

NALOXONE KIT

- Naloxone is an **opioid antagonist** that displaces opioids at their receptor sites
- Able to **temporarily reverse the effects of an overdose** within minutes
- Naloxone is available as either a **nasal spray or injection**
- It can be obtained without a prescription from most retail pharmacies (offered free of charge in many provinces)
- Pharmacy staff should offer naloxone kits to any patient who may be at risk of overdose or anyone caring for someone who may be at risk

WHAT ARE THE SIGNS OF AN OPIOID OVERDOSE?

- Pale and clammy skin
- Extreme drowsiness or loss of consciousness
- Unable to be woken up by shouting, shaking of shoulders or touch
- Breathing that is very slow, irregular or has stopped
- Slowed or no heartbeat
- Pinpoint pupils



PROFESSIONAL SERVICES

NALOXONE KIT EXAMPLE QUESTION

CASE STUDY

RR is a regular patient who comes to your pharmacy to pick up his chronic back pain medication. You notice that he is now dropping off a new prescription for a dose increase on his Hydromorph Contin® along with a renewal prescription for his Lenoltec #3. Upon clinically verifying the prescriptions, you notice naloxone kit has never been billed on his profile. Upon pick up, you offer RR a naloxone kit and he accepts, how do you counsel him on its use?

SPRAY

INJECTION

PROFESSIONAL SERVICES

PAXLOVID® PRESCRIBING

150 mg Nirmatrelvir/100 mg ritonavir (Paxlovid) is an oral antiviral medication indicated for the treatment of patients with mild to moderate COVID-19 who are at high risk of disease progression, including hospitalization or death

- Must be initiated **within five days of symptom onset** to be effective
- Must have a positive COVID-19 result
- Not used for pre- or post-exposure prophylaxis or for treatment of severe/critical COVID-19 illness
- Regular dosage regimen: 2 tablets of nirmatrelvir and 1 tablet of ritonavir BID for 5 days.
 - Renal dosing is available for those with renal impairment
- Ritonavir is a CYP3A4 inhibitor → check drug interactions!

PROFESSIONAL SERVICES

COMPLIANCE PACKS

- Method of dispensing medication in **weekly packs** with **divided dosing** for medication at breakfast, lunch, dinner and bedtime
- Dispensed every 1-4 weeks in community pharmacy on a cycle (varies from one pharmacy to another)



PROFESSIONAL SERVICES

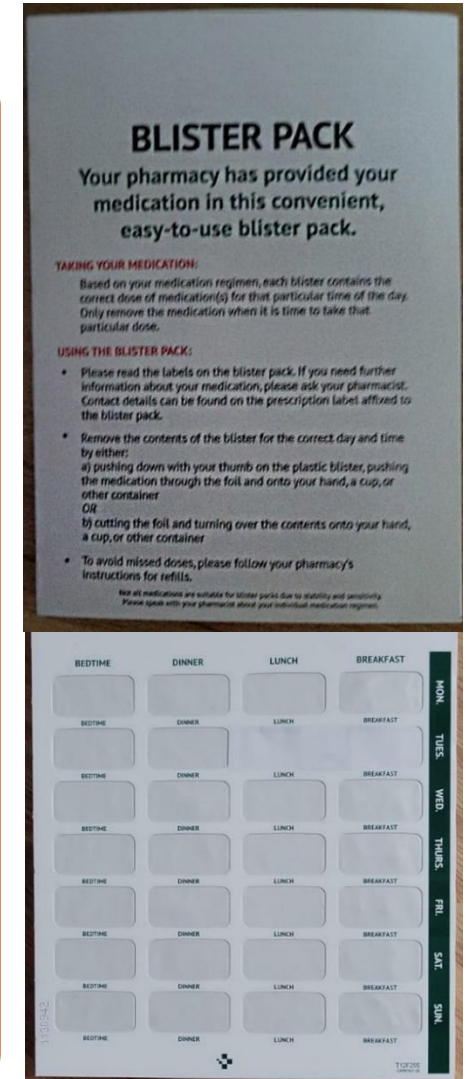
COMPLIANCE PACKS

Advantages

- Useful for patients who take many medications at different times
- Helpful for patients who may forget if they have taken their medication, reduces risk of double dosing, missing a dose
- Convenient method to store medication, especially when travelling
- May be easier for patients who have dexterity issues

Disadvantages

- If the patient's medication changes, the entire 4- week blister pack may need to be re-dispensed
- Drugs are not protected from light
- Blister packs are not child-proof
- Difficult to explain to patients how to space out morning medication if they take medication that needs to be spaced out throughout the morning. For example, a patient who takes lansoprazole, alendronate and levothyroxine would need to take all of these in the morning but spaced apart by at least half an hour. In order to do this, the patient would still have to be able to identify which medication is which



PROFESSIONAL SERVICES

COMPLIANCE PACK EXAMPLE

CASE STUDY

- YS comes to your pharmacy and wants to be set up on a blister pack. He says he is having trouble remembering to take his medications and that a blister pack might help with this. He gives you a current list of his medications.
 - Escitalopram 10 mg at night, rosuvastatin 10 mg at night, risedronate 35 mg once weekly on Monday mornings, Centrum® multivitamin once daily, perindopril 4 mg in the morning along with olmesartan and hydrochlorothiazide 40/25 mg combination pill

	Morning (8AM)	Noon (12PM)	Supper (5PM)	Bedtime
Escitalopram 10mg				
Rosuvastatin 10mg				
Risedronate 35mg				
Centrum® multivitamin				
Perindopril 4mg				
Olmesartan and HCTZ 40/25mg				

TAKEAWAY

- Prescription drug coverage in Canada includes **public** (provincial/federal) and **private** (e.g. workplace benefits, out-of-pocket) options
- **Formularies** list drugs approved for coverage after a Health Technology Assessment based on safety, efficacy, and cost
- **Total prescription cost = drug ingredient cost + markup + dispensing fee**
- Drug Use Evaluation helps **improve** patient outcomes and **reduce** medication-related problems
- **Professional services** include medication reviews, minor ailment prescribing, immunizations, smoking cessation and more

EXTRA INSURANCE CALCULATIONS PRACTICE QUESTION

Question 1: You receive a prescription for a prednisone taper with the following directions:

Rx

Dispense in prednisone 5mg tablets

Take 6 tablets (=30mg) daily for 1 week, then

Take 5 tablets (=25mg) daily for 1 week, then

Take 4 tablets (=20mg) daily for 1 week, then

Take 3 tablets (=15mg) daily for 1 week, then

Take 2 tablets (=10mg) daily for 1 week, then

Take 1 tablet (=5mg) daily for 1 week, then stop

If each prednisone tablet costs \$0.30, calculate how much will the patient pay for the total quantity of the medication?

- Pharmacy dispensing fee is \$10
- The patient has no private plan

EXTRA INSURANCE CALCULATIONS PRACTICE QUESTION

Question 2: Protopic® Ointment costs \$1.70 for 1 gram. The pharmacy charges a markup of 6%. You receive the following prescription:

Rx

Protopic® 0.01% ointment to be applied to affected area(s) 3 times daily as needed

Mitte = 60g

Repeat = 0

Calculate the cost of the prescription if the patient has no insurance.

- Dispensing fee = \$7

EXTRA INSURANCE CALCULATIONS PRACTICE QUESTION

Question 3: RK comes to your pharmacy and asks to fill a prescription for amoxicillin 300mg 3 times daily for 7 days. Upon checking your pharmacy system, you realize that amoxicillin is not available in this strength. Upon patient approval and adaptation regulations in your province, you decide to dispense the amoxicillin as a liquid suspension. Patient says she has no insurance. Pharmacy Dispensing Fee is \$9.

Patient mentions that she wants to take the smallest amount of liquid possible. Calculate her total cost for the prescription.

Available strengths of liquid amoxicillin:

- 125mg/5ml → 100 ml bottle costs \$16.80, 150ml bottle costs \$24.60
- 250mg/5ml → 100 ml bottle costs \$17.50, 150ml bottle costs \$25.48

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CHANGE LOG

August 2025

- Slide 6: Changed to "Yukon: Pharmacare program"
- Slide 29: Updated to "focus is mainly on cost"
- Slide 39: Added "offered free of charge in many provinces"
- Slide 45: Added 'Takeaway' slide
- Formatting and grammatical changes made throughout

November 2025

- Slide 3: Added RCMP, P & T, MAC, CDE, DUE and BCGP to legend. Removed CADTH from legend
- Slide 6: Added FYI
- Slide 7: Removed FYI
- Slide 17: Removed FYI

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 23: Added to show question only without the answer
- Slide 27: Added slide with answer to question 3